The above table provides a neighbourhood host reachability status as well. It classifies the reachability in four classes – valid permanent connection, valid but not further tested, reachable but with the reachable timeout limit, and suspicious. With all these, it is possible to get a good understanding of your neighbourhood to investigate the performance of the network, if required.

Services and socket connectivity

To understand which services are running within your server and expecting outside hosts to cater their services to them, you need to go through the currently running services and examine their status. Naturally, these include DNS, NFS, Web services and other distributed services within the network. A close look at this data can help a network analyst or security expert to easily identify the possible sources of performance bottlenecks and external attacks. The information is based upon the ports that are open and in the *listening* state, waiting for a client's connections, or to those that are already open and communicating with a client.

Sockets

A socket is the handshaking protocol that provides connection-oriented communication between a server and all the hosts within its network. It has a well-defined flow of events. In a connection-oriented client-to-server model, the socket on the server process awaits requests from a client. The server first binds an address and a port to the socket that a client can use to find the server. When there is a definite address and port combination for a service, the server waits for clients to request the service through its own address and port combination. The server performs the client's request and sends the reply to the client.

Proto	Recv-Q	Send-Q	Local address	Foreign address	State	PID/Program name
tcp	0	0	0.0.0.0:10000	0.0.0.0:*	LISTEN	40979/perl
tcp	0	0	0.0.0.0:80	0.0.0.0:*	LISTEN	2459/httpd
tcp	0	0	127.0.0.1:5910	0.0.0.0:*	LISTEN	14967/Xvnc
tcp	0	0	127.0.0.1:3350	0.0.0.0:*	LISTEN	1568/xrdp-sesman
tcp	0	0	0.0.0.0:22	0.0.0.0:*	LISTEN	1562/sshd
tcp	0	0	127.0.0.1:25	0.0.0.0:*	LISTEN	42415/sendmail: acc
tcp	0	0	0.0.0.0:443	0.0.0.0:*	LISTEN	2459/httpd
tcp	0	0	172.17.0.11:443	49.15.183.212:28612	SYN_RECV	-
tcp	0	0	172.17.0.11:443	23.16.137.141:60054	SYN_RECV	-
tcp	0	0	0.0.0.0:3389	0.0.0.0:*	LISTEN	1570/xrdp
tcp	0	0	172.17.0.11:443	103.16.202.222:58515	TIME_WAIT	-
tcp	0	224196	172.17.0.11:443	49.15.163.23:18485	ESTABLISHED	4251/httpd

Figure 3: Tabular outputs of *netstat -antplF* command.

Proto	Recv.Q	Send.Q	Local.Address	Foreign.Address
tcp :195	Min. :0	Min. : 0	172.16.200.88:443:76	172.16.200.89:3306:45
tcp6 : 20	1st Qu.:0	1st Qu. : 0	172.17.0.11:443 :25	0.0.0.0:* : 26
	Median :0	Median : 0	172.16.200.88:22 : 8	:::* : 20
	Mean :0	Mean : 302	:::22 : 3	172.17.0.11:3306:10
	3rd Qu. :0	3rd Qu. : 0	:::7937 : 3	172.16.101.192:62835: 2
	Max. :0	Max. : 282240	:::7938 : 3	172.16.101.192:62839:2
			(Other) :97	(Other) :110
State	PID.Program.name			
ESTABLISHED : 34	:133			
FIN_WAIT1 : 2	96251/nsrexecd : 10			
FIN_WAIT2 : 16	1637/nsrexecd : 5			
LISTEN : 46	1247/sshd : 4			
SYN_RECV : 2	29986/httpd : 3			
TIME_WAIT :115	113426/sshd: root@n: 2			

Figure 4: R summary report of the *netstat* status log